

$^{35}\text{Cl}(\text{n},\text{n}'\gamma)$ 1989Ge09,1984El12,1966Ni03

1984El12,1989Ge09: fast neutrons of diameter 26 mm were produced from the IRT-2000 reactor at the Institute for Nuclear Research and Nuclear Power of the Bulgarian Academy of Sciences in Sofia. Two ^{35}Cl targets were placed at 45° and 135° with a distance of 70 cm apart. γ rays were detected using a Ge(Li) detector. Measured 4 E_γ . Deduced lifetimes for 4 levels using the Doppler Shift Attenuation Method (DSAM) and transition strengths.

1966Ni03: 2.5-4.2-MeV neutrons were produced via the T(p,n) reactions with protons from the university of Kentucky 6-MeV accelerator. Targets were a cylinder of CCl_4 and a 5.6-cm-diam sphere of C_2Cl_6 . γ rays were detected using a 6.35-cm by 6.35-cm NaI(Tl) detector and 7.5-cc Ge(Li) detector. Measured 7 E_γ and I_γ , and 3 $\gamma(\theta)$. Deduced 6 levels and 3 spins.

1971Fr05: neutrons up to 10 MeV were produced from a 5-Ci Pu-Be neutron source and bombarded table salt targets. γ rays were detected using a Ge(Li) detector. Measured 2 E_γ and deduced 2 levels.

Others: 1955To32, 1977Ko19, 1983Gl07, 1992Be60 (cross sections).

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\#$	$T_{1/2}^\ddagger$
0	$3/2^+$	
1219.0 10	$1/2$	139 fs 56
1762.1 10	$5/2$	
2646.1 10	$7/2$	187 fs 62
2694.1 10		43 fs 6
3002.1 10		50 fs 9
3163.2 10		

† From a least-squares fit to γ -ray energies.

‡ From 1984El12 and 1989Ge09 using DSAM.

$^\#$ From $\gamma(\theta)$ in 1966Ni03, except for the ground state from the Adopted Levels.

 $\gamma(^{35}\text{Cl})$

E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
(882)	2646.1	$7/2$	1762.1	$5/2$	Known γ but not directly observed in this dataset. $B(\text{M1})(\text{W.u.})=0.014$ 1, $B(\text{E2})(\text{W.u.})=4.3$ 14 (1989Ge09).
(930)	2694.1		1762.1	$5/2$	Known γ but not directly observed in this dataset. $B(\text{M1})(\text{W.u.})=0.087$ 11, $B(\text{E2})(\text{W.u.})=3.1$ 4 (1989Ge09).
1219	1219.0	$1/2$	0	$3/2^+$	$B(\text{M1})(\text{W.u.})=0.085$ 34, $B(\text{E2})(\text{W.u.})=3.6$ 14 (1989Ge09).
1762	1762.1	$5/2$	0	$3/2^+$	E_γ : from 1966Ni03.
					$A_0=+0.942$ 8, $A_2=-0.148$ 23 (1966Ni03).
2646	2646.1	$7/2$	0	$3/2^+$	$A_0=+1.078$ 5, $A_2=+0.165$ 12 (1966Ni03).
					$B(\text{E2})(\text{W.u.})=3.1$ 11 (1989Ge09).
2694	2694.1		0	$3/2^+$	$B(\text{M1})(\text{W.u.})=0.020$ 3, $B(\text{E2})(\text{W.u.})=0.3$ 1 (1989Ge09).
3002	3002.1		0	$3/2^+$	$B(\text{M1})(\text{W.u.})=0.016$ 3, $B(\text{E2})(\text{W.u.})=0.4$ 1 (1989Ge09).
3163	3163.2		0	$3/2^+$	E_γ : from 1966Ni03.

³⁵Cl(n,n'γ) 1989Ge09,1984El12,1966Ni03

Legend

Level Scheme

-----► γ Decay (Uncertain)

